SS-P-155d

FEBRUARY 7, 1961

SUPERSEDING
Int. Fed. Spec. SS–P–00155c(GSA–FSS)
October 20, 1960 and
Fed. Spec. SS–P–155a
April 11, 1958

FEDERAL SPECIFICATION

PENCIL AND LEAD, ELECTROGRAPHIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers a wood-cased electrographic lead pencil, and electrographic leads for mechanical pencils, used in scoring and mark sensing on electrographic equipment (see 6.1).

1.2 Classification.

1.2.1 *Type.*—The electrographic lead pencil, and electrographic leads for mechanical pencils, shall be of but one type.

1.2.2 *Length of leads (for mechanical pencils).*—(See 3.5).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

GG–S–236—Sharpeners, Pencil.

SS–P–201—Pencil, Nonmechanical, Colored Lead.

UU–P–31 — Paper; General Specifications and Methods of Testing.

ZZ–E–661 — Erasers, Rubber and Rubber Substitute.

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–636 — Box, Fiberboard.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–L–10547 — Liners, Case, Waterproof.

570402/61/243

FSC 7510

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—Standards of the Technical Association of the Pulp and Paper Industry, referenced herein, form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

(Copies of T.A.P.P.I. Standards may be obtained from the Technical Association of the Pulp and Paper Industry, 155 East 44th Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Bid sample.—When specified, each bidder shall submit with his bid six samples of the wood-cased electrographic pencils and/or leads for mechanical pencils he proposes to furnish for the purpose of performance testing to assure satisfactory scoring and mark sensing performance prior to award (see 6.2). Pencils or leads furnished under contract shall be at least equal to the respective bid samples in regard to these characteristics.

3.2 Certification. — When specified, each bidder shall submit with his bid a certificate, including test results from a recognized laboratory acceptable to the procuring agency, showing that the wood-cased electrographic pencils and/or leads for mechanical pencils he proposes to furnish comply with the electrical resistance requirements in 3.4.3 and 3.5 and the applicable test methods of this specification (see 6.2).

3.3 Materials.—Materials shall be as specified hereinafter.

3.4 Pencils, wood-cased.

3.4.1 *Construction.* — Wood-cased electrographic pencils shall be round or hexagonal shaped. The lead shall be in one piece and shall be bonded to the wood casing the entire length of the pencil. Both halves of the wood casing shall be of uniform structure and securely glued together the entire length. The lead shall be centered the length of the casing within a tolerance of 0.025 inch. Each pencil shall be provided with an eraser securely held by a metal band.

3.4.2 *Casing.* — The casing of the wood-cased electrographic pencil shall be made of soft, well-seasoned, straight-grained wood, shall not split or splinter and shall be readily cut with a pencil sharpener conforming to type II of Federal Specification GG–S–236 and the wood shall take a smooth, even finish. The wood casing shall be finished with blue or black lacquer.

3.4.3 *Lead.*—The lead shall be made from a black graphite mixture and shall produce smooth, even, and uniform writing (see 4.5.1.1). The diameter of the lead shall be 0.100 plus or minus 0.006 inch, and shall have a breaking strength of 2 pounds when tested as specified in 4.5.1.2. The lead shall produce marks having an average electrical resistance of 190 to 380 ohms when tested as specified in 4.5.1.3.

3.4.4 *Eraser.*—The eraser shall conform to the requirements of Federal Specification ZZ–E–661, type I, grade B.

3.4.5 *Length.*—Pencils shall be at least 6¾ inches in length, not including the metal band and eraser.

3.4.6 *Identification marking.*—Each wood-cased electrographic pencil shall be legibly stamped with the name of the manufacturer and designation indicating the kind of pencil, such as "electrographic," "mark sensing," or "test scoring."

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3.5 Mechanical pencil leads. — Leads for mechanical pencils shall be of the same material as that specified in 3.4.3 for wood-cased pencils and shall produce smooth, even, and uniform writing (see 4.5.1.1). Each electrographic lead shall be 0.046 to 0.047 inch in diameter. Unless otherwise specified, leads shall be 4 inches or 1 3/8 inches long, as specified (see 6.2). A tolerance of plus or minus 1/32 inch will be permitted in length of leads. Electrographic lead shall have a breaking strength of 27 inch grams when tested as specified in 4.5.2.2. The lead shall produce marks having an average electrical resistance of 300 to 600 ohms when tested as specified in 4.5.2.3. The 4-inch leads may be marked 3/8 plus or minus 1/32 inch from one end of the lead with a contrasting and readily perceptible color to indicate that the lead is nearly exhausted. Such marking shall not adversely affect the writing qualities of the lead when tested as specified in 4.5.2.1.

3.6 Workmanship.—Electrographic pencils shall be smoothly finished and shall be free from splits, splinters, blisters and any other imperfections which affect the appearance or which might impair the serviceability. Leads for mechanical pencils shall be in accordance with the quality requirements of this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier's responsibility. — Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.)

4.1.1 *Inspection and test of preparation for delivery requirements.* — The preservation, packaging, packing, and marking of electrographic pencils, and leads for mechanical pencils, shall be subject to inspection and test by the Government inspector to determine compliance with the requirements of section 5 of this specification. The contractor shall provide the necessary facilities and supplies for inspections and tests made prior to final acceptance.

4.2 Lot.—A lot shall consist of all electrographic pencils, or leads of one size for mechanical pencils, offered for delivery at one time.

4.3 Sampling.—Sampling for examination and tests shall be conducted in accordance with Military Standard MIL–STD–105.

4.4 Examination.—Samples of pencils and leads shall be inspected in accordance with the classification of defects shown in table I. The Acceptable Quality Level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Tests.—The following tests shall be conducted on samples of pencils and leads selected in accordance with Military Standard MIL–STD–105. Inspection Level L1. Any sample which fails to meet the test requirements shall be cause for rejection of the lot.

4.5.1 *Wood-cased pencils.*

4.5.1.1 *Writing.* — Each sample electrographic pencil shall be sharpened in a pencil sharpener as specified in 3.4.2 to a fine point at seven different points, at least 1/2 inch apart, and tested at each point by drawing a line equivalent to 24 inches under normal writing pressure and the lines shall show uniformity of color and evenness. When drawing the line, the lead shall work smoothly without

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TABLE I.—*Classification of defects*

Defects	Category		Applicable reference paragraph
	Major	Minor	
Wood-cased electrographic pencils			
Casing not securely glued together	X		3.4.1
Eraser loose	X		3.4.1
Lead:			
Not centered	X		3.4.1
Not bonded	X		3.4.1
Diameter not as specified	X		3.4.3
Wood casing does not take a smooth, even finish when sharpened	X		3.4.2
Finish:			
Not with blue or black lacquer	X		3.4.2
Imperfections affecting:			
Appearance		X	3.6
Serviceability	X		3.6
Length of pencil not as specified	X		3.4.5
Identification marking not as specified or illegible	X		3.4.6
Mechanical pencil leads			
Length not as specified	X		3.5
Diameter not as specified	X		3.5
Colored marking (when furnished on 4-inch leads) not located as specified		X	3.5

grittiness. Lines shall be drawn on test scoring paper as described in 4.5.1.3.2.

4.5.1.2 *Breaking strength.*—The breaking strength of the electrographic lead shall be determined as described in Federal Specification SS–P–201. The pencils shall be sharpened to points having diameters of 0.050 plus or minus 0.001 inch. Crumbling, breaking, splintering, or chipping of leads under pressure less than specified in 3.4.3 shall be considered cause for rejection.

4.5.1.3 *Electrical resistance of lead.*

4.5.1.3.1 *Humidity and temperature.*—The test shall be conducted at a relative humidity of 50 plus or minus 2 percent and a temperature of 73° plus or minus 3.5°F.

4.5.1.3.2 *Scoring paper.*—Scoring paper used in the test shall meet the requirements specified in table II.

4.5.1.3.3 *Point of pencil.* — The exposed pencil lead shall be turned down to a uniform diameter of 0.062 inch. Before test marks are made, the end of the lead shall be flattened to a plain surface at exact right angles to the axis of the pencil.

4.5.1.3.4 *Making of test marks.*—In making marks, the pencil shall be held firmly by a mechanical device at right angles to the paper with a pressure of 2 pounds, 11 ounces. To insure uniformity of markings, the paper may be held against the pencil on a cylindrical drum measuring 1⅞ inches to 2 inches in diameter. The mark shall be made by three superimposed strokes of the pencil. The mark shall be approximately 4 inches long, and the successive strokes can be made without lifting the pencil from the paper. In making the mark, the pencil shall be moved over the paper at a speed of 25 feet per minute or the pencil may be held stationary and the paper moved at a speed of 25 feet per minute. The marks shall be drawn across the grain of the paper on the felt side.

4.5.1.3.5 *Sensing the marks.*—The contact for taking readings shall consist of five parallel phosphor-bronze blades 0.008 inch thick, 0.100 inch wide, spaced 0.015 inch apart. The two outer blades and the center blade shall be connected in parallel to form one side of the line. The remaining two blades shall be connected in parrallel to form the other side

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TABLE II.—*Scoring paper*

Characteristics	Test Method	
	Fed. Spec. UU–P–31	T.A.P.P.I.
	Number	*Number*
Stock: 100 percent bleached chemical wood pulp	300	T 401 m
Basis weight: 17 x 22 – 500, 20 pounds plus or minus 1 pound	110	T 410 m
Thickness: 0.0045 to 0.0050 inch	173	T 411 m
Bursting strength: Average not less than 20 pounds per square inch	112	T 403 m
Stiffness (Taber V–5 Model 150) Machine direction: Average not less than 2.5 units. Cross direction: Average not less than 1.3 units.	---	T 451 m
Ash: 7.0 percent, maximum	202	T 413 m
Opacity: 85.0 percent contrast ratio, minimum	151	T 425 m
Moisture expansivity Machine direction: 0.07 percent, maximum. Cross direction: 0.20 percent, maximum.	---	T 447 m
Moisture content: 3.5 to 5.5 percent	240	T 412 m
Insulation resistance (at 80 percent relative humidity): Average of 10 specimens shall be not less than 50 megohms	*	*
Finish and tooth	**	**

* Resistance meter, SIE Model C–6, manufactured by the Southwestern Industrial Electronics Company, or any other similar to or equal meter capable of measuring a resistance of 50 megohms or higher, may be used for the insulation resistance test. Any laboratory method of test will be satisfactory.

** Test scoring paper shall conform to standard sample in surface finish and tooth to insure electrical conductivity of marks made with an electrographic pencil (see 6.3).

of the line. The terminals of the two sets of blades shall be connected to an ohmmeter having a voltage range from 1½ to 6 volts. The blades shall be placed against the pencil mark with a pressure of 3 pounds, and the resistance measured between the two sets of blades. The paper shall be backed by a hard-surfaced rubber pad to assure a good contact between the contact blades and the pencil mark.

4.5.1.3.6 *Determination of average electrical resistance.*—The electrical resistance of each mark shall be measured in five places within ½ inch of the center of the mark. The five readings obtained shall be averaged out to arrive at the average electrical resistance for each pencil (190 to 380 ohms). When there is evidence of surface imperfections on the paper that will not permit the pencil to produce a uniform deposit of lead, the reading in this area shall be waived and not be included in the average.

4.5.1.4 *Erasers.*—Erasers furnished with electrographic pencils shall be tested for conformance with the requirements of Federal Specification ZZ–E–661, type I, grade B.

4.5.2 *Mechanical pencil leads.*

4.5.2.1 *Writing.*—Tests shall be conducted as described in 4.5.1.1, as applicable, except that the lines shall be drawn from each end of the lead and 4-inch leads furnished with color markings shall be tested at the point of such marking.

4.5.2.2 *Breaking strength.*—The breaking strength of the electrographic lead shall be based upon the load in grams required to break the lead when the load is applied so as to impose a bending moment upon lead. A load shall be applied to the lead to determine if it meets a minimum breaking strength. For example, a length of lead held firmly at one end and subjected to a load of 160 grams applied ¼ inch from the support meets the minimum breaking load requirement if the lead does not break since 160 grams x ¼ inch equals 40 inch grams.

4.5.2.3 *Electrical resistance of lead.*—The test shall be conducted as described in 4.5.1.3

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except the lead shall be 0.046 in diameter, markings made at a pressure of 1 pound, 8 ounces, and average electrical resistance 300 to 600 ohms.

5. PREPARATION FOR DELIVERY

5.1 Federal Standard No. 102 (for civil agencies only).—The definitions and applications of the levels of packaging and packing protection shall be in accordance with Federal Standard No. 102.

5.2 Packaging.—Wood-cased electrographic pencils, and leads for mechanical pencils, shall be packaged level A, B or C, as specified in the invitation for bids, contract or order (see 6.2).

5.2.1 *Levels A and B.*

5.2.1.1 *Pencils.* — Pencils shall be unit-packaged 1 dozen to a sleeve or other suitable container commonly used for the purpose. Six unit packages shall be intermediate-packaged in a folding or set-up box conforming to Federal Specifications PPP–B–566 and PPP–B–676, respectively.

5.2.1.2 *Leads for mechanical pencils.*

5.2.1.2.1 *Military agencies.* — Leads for mechanical pencils shall be unit-packaged 72 each in a box, tube or other suitable container commonly used for the purpose. Six unit packages shall be intermediate-packaged in a folding or set-up box conforming to Federal Specifications PPP–B–566 and PPP–B–676, respectively.

5.2.1.2.2 *Civil agencies.*—Unless otherwise specified, leads for mechanical pencils shall be unit- and intermediate-packaged as specified in 5.2.1.2.1.

5.2.2 *Level C.*—Electrographic pencils, and leads for mechanical pencils, shall be packaged in accordance with standard commercial practice.

5.3 Packing.—Pencils and leads, packaged as specified in 5.2, shall be packed level A, B or C, as specified in the invitation for bids, contract or order (see 6.2).

5.3.1 *Level A.*—Forty intermediate boxes of pencils (2880 pencils) shall be packed in a snug-fitting fiberboard box conforming to Federal Specification PPP–B–636, class 2, V3c or V3s. Shipping containers shall be provided with waterproof case liners conforming to class 2 of Military Specification MIL–L–10547 and sealed in accordance with the Appendix thereto. Leads shall be packed in the containers specified for electrographic pencils, gross weight not to exceed the weight limitations of the box specification.

5.3.2 *Level B.*—Unless otherwise specified, forty intermediate boxes of pencils (2880 pencils) shall be packed in a snug-fitting corrugated or solid fiberboard box conforming to Federal Specification PPP–B–636, type I or II, class 1. Burst test of fiberboard used in the box construction, unless otherwise specified, shall be not less than 175 p.s.i. Closure shall be in accordance with Federal Specification PPP–B–636. Leads shall be packed in the containers specified for electrographic pencils, gross weight not to exceed the weight limitations of the box specification.

5.4 Marking.

5.4.1 *Civil agencies.*— In addition to any special marking specified in the contract or order, packages and shipping containers shall be marked in accordance with Federal Standard No. 123. Unless otherwise specified, unit packages shall be marked with the Federal Stock Number, contractor's name and address, contract or order number, and the symbol-number of this specification. Intermediate packages, unless otherwise specified, shall be marked on one end with the Federal Stock Number, Federal Item Identification, including specification symbol-number and item name, quantity of units contained (e.g., SIX DOZEN), manufacturer's name or con-

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tractor's, if different, and the contract or order number.

5.4.2 *Military agencies.*—In addition to any special marking specified in the contract or order, intermediate and shipping containers shall be marked in accordance with Military Standard MIL–STD–129. Unit packages shall be marked in accordance with manufacturer's commercial practice and shall include name of manufacturer and designation of pencil or lead, such as "electrographic," "mark sensing," or "test scoring."

5.5 Standard pack (civil agencies).—The standard pack requirements for the items covered by this specification shall be as specified in 5.2.1 (level B), 5.3.2 and 5.4.1. (See 6.4.)

6. NOTES

6.1 Intended use. — The electrographic pencils, and leads for mechanical pencils, covered by this specification are intended for writing conductive markings on test scoring papers and mark sensing cards used on electrographic equipment.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following (see 6.4):

(*a*) Title, symbol-number, and date of this specification.
(*b*) When a bid sample or a certificate of compliance, or both, is required (see 3.1 and 3.2).
(*c*) Whether wood-cased electrographic pencils, or leads for mechanical pencils are required (see 3.4 and 3.5).
(*d*) Data, as required, regarding examination and tests; and inspection responsibility for civil agency procurement if other than as specified (see 4.1).
(*e*) Level of packaging and level of packing and any special marking required (see 5.2, 5.3, and 5.4).
(*f*) Any packaging, packing and/or marking options desired (see 5.2.1.2.2, 5.3.2 and 5.4.1).

6.2.1 *Military standards.*—Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.3 Standard sample (test scoring paper).—Standard sample of test scoring paper (par. 4.5.1.3.2, table II) will be available for information or inspection at the address of the procuring agency.

6.4 The standard pack requirements in 5.2.1 (level B), 5.3.2, and 5.4.1 are intended for use in procurements of Stores Stock replenishments. Procuring officers should use the standard pack provisions when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

6.5 Transportation data.—Transportation descriptions and minimum weights applicable to this commodity are:

Rail—Pencils
Pencil, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:
Pencils not otherwise indexed.
Motor volume minimum weight 24,000 pounds.

Rail—Leads
Pencil leads.
Carload minimum weight, 30,000 pounds.

Motor
Pencil leads.
Truckload minimum weight (w) 30.6 pounds, (w) subject to Rule 34, National Motor Freight Classification.

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MILITARY INTERESTS:

Army—Quartermaster Corps
Navy—Bureau of Supplies and Accounts
Air Force.

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